

SMARTBRANCH 360

Network Design Document

Build a Secure Branch Office Network and Prove It Works

Program	Cisco Virtual Internship 2026
Track	Networking + Cisco Packet Tracer + Python
Project	SmartBranch 360
Document Type	Design and Implementation Architecture
Prepared	August 2026

Design baseline: VLAN segmentation, router-on-a-stick, DHCP/DNS, Guest Wi-Fi, NAT/PAT, ACL isolation, management-only SSH, structured testing, fault injection, and Python assurance.

1. Project Overview

SmartBranch 360 is a small branch-office network designed and implemented in Cisco Packet Tracer. The project combines practical switching, routing, network services, security controls, troubleshooting, and lightweight Python-based configuration assurance. The branch is divided into separate Employee, Guest, Server, and Management networks so that traffic can be controlled according to business role rather than allowing every endpoint to share one flat LAN.

The completed topology uses one Cisco 2911 router, two Cisco 2960-24TT switches, one wireless access point, an internal server, a simulated Internet server, wired employee endpoints, two wireless guest laptops, and a dedicated administration PC. Inter-VLAN routing is performed on R1-BRANCH using 802.1Q router subinterfaces. The router also provides the branch edge toward the simulated 203.0.113.0/24 external network.

Project principle: A working configuration is not treated as complete until the required behavior is tested, a fault can be diagnosed from evidence, and the Python checker independently validates important design requirements.

2. Problem Statement

A company is opening a new branch office that needs wired employee access, guest Wi-Fi, an internal service host, Internet reachability, and secure administration of network devices. A flat network would make access control difficult, increase the impact of configuration mistakes, and allow guest traffic to reach sensitive internal resources. The branch therefore requires a structured design that separates users by function and applies explicit security policy between those functions.

The project must also address operational reliability. Configuration errors such as a wrong default gateway, a missing VLAN on a trunk, an incorrect DHCP option, an ACL rule that blocks DNS, or a missing NAT role can break selected services while the rest of the network appears healthy. SmartBranch 360 addresses this by combining normal-operation tests, intentional fault injection, command-line evidence, and an automated Python assurance tool.

3. Project Objectives

- Create a clearly labeled Packet Tracer branch topology using one router, two switches, a wireless AP, servers, and at least eight client endpoints.
- Segment the branch into VLAN 10 Employee, VLAN 20 Guest, VLAN 30 Server, and VLAN 99 Management.
- Assign a separate /24 IPv4 subnet and a predictable default gateway to each VLAN.
- Implement inter-VLAN routing using router-on-a-stick with 802.1Q encapsulation.
- Provide DHCP and DNS reachability, guest wireless connectivity, and NAT/PAT for external access.
- Block Guest users from Employee, Server, and Management networks while preserving required external access.
- Restrict SSH management so that only the administration host at 10.10.99.10 is accepted as the management source.
- Validate healthy operation with endpoint tests and Cisco show commands.
- Inject and repair at least five representative configuration faults using evidence-based troubleshooting.
- Use a Python checker to compare captured Cisco outputs with the documented SmartBranch 360 requirements and report PASS, WARN, or FAIL findings.

4. Network Architecture

4.1 Physical Topology

The physical topology follows a compact hierarchical branch design. R1-BRANCH connects downward to SW1, which acts as the main access/distribution switch for the branch. SW1 also connects to SW2 through an 802.1Q trunk, extending all four VLANs to the second access area. AP-GUEST is connected to SW2 on the Guest VLAN. R1-BRANCH connects separately to INTERNET-SRV on the simulated external network.

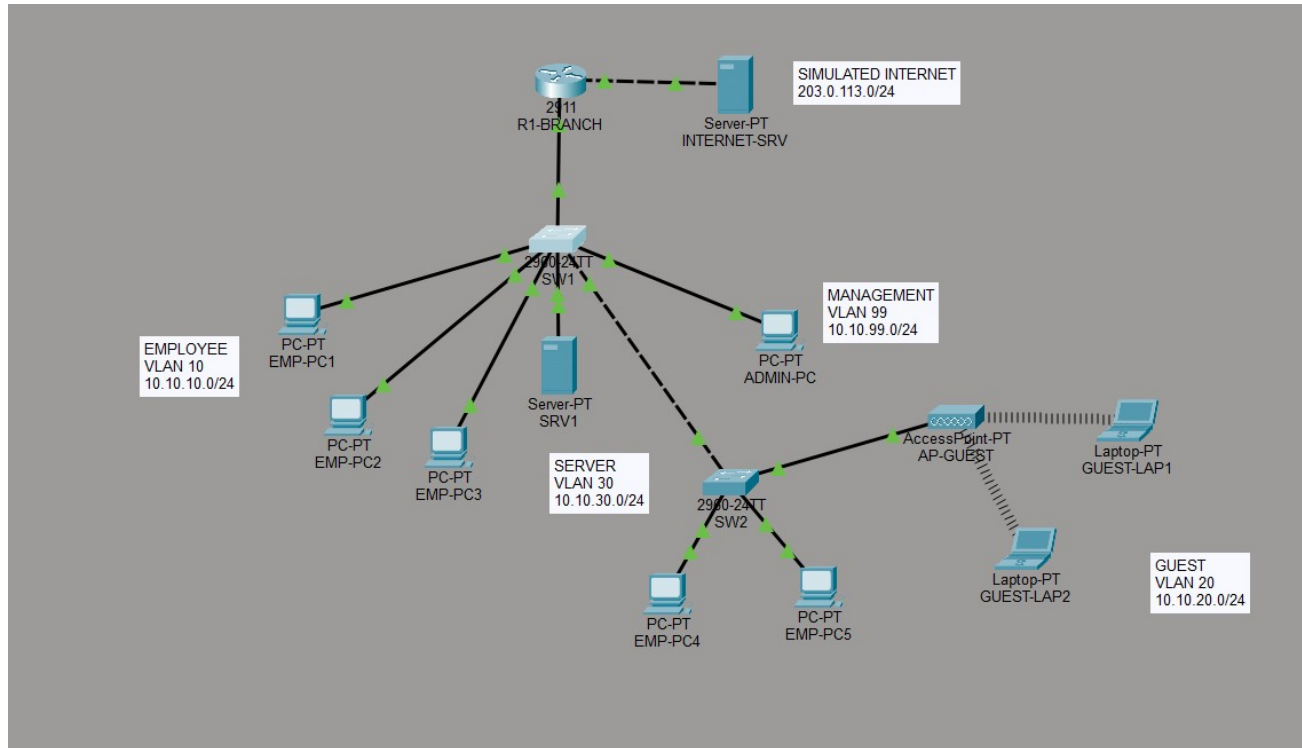


Figure 1. SmartBranch 360 Packet Tracer topology (submitted topology screenshot).

4.2 Logical Topology

Logically, the branch is divided into four Layer-2 broadcast domains. Access links place endpoints in exactly one VLAN, while the router-facing and inter-switch links carry multiple VLANs as trunks. R1-BRANCH terminates one logical Layer-3 interface per VLAN and therefore becomes the default gateway for every branch subnet. The Guest VLAN is routed through an inbound ACL before it can reach internal or external destinations.

Logical Zone	VLAN	Subnet	Gateway	Primary Purpose
Employee	10	10.10.10.0/24	10.10.10.1	Wired employee endpoints and normal business access
Guest	20	10.10.20.0/24	10.10.20.1	Wireless visitor access with internal isolation
Server	30	10.10.30.0/24	10.10.30.1	Internal server and DNS service
Management	99	10.10.99.0/24	10.10.99.1	Administration and device management

Logical Zone	VLAN	Subnet	Gateway	Primary Purpose
External	-	203.0.113.0/24	R1 G0/1 = 203.0.113.1	Simulated Internet segment

4.3 Device Inventory

Device / Group	Model / Type	Qty.	Role / Placement
R1-BRANCH	Cisco 2911 Router	1	Inter-VLAN routing, DHCP, NAT/PAT, ACL enforcement, external edge
SW1	Cisco 2960-24TT	1	Primary access/distribution switch; router trunk and SW2 trunk
SW2	Cisco 2960-24TT	1	Secondary access switch; employee and Guest AP connectivity
AP-GUEST	AccessPoint-PT	1	Wireless access for VLAN 20 Guest
SRV1	Server-PT	1	Internal server at 10.10.30.10; internal DNS service host
INTERNET-SRV	Server-PT	1	Simulated external service host at 203.0.113.10
EMP-PC1 to EMP-PC5	PC-PT	5	Employee VLAN 10 clients
ADMIN-PC	PC-PT	1	Management host at 10.10.99.10; authorized SSH source
GUEST-LAP1 / GUEST-LAP2	Laptop-PT	2	Wireless Guest VLAN 20 clients

5. VLAN Design

VLAN segmentation is the foundation of the SmartBranch 360 security and addressing design. The same VLAN IDs and names are created on both SW1 and SW2, and all required VLANs are carried across the intended trunk links. This makes user placement independent of switch location while still preserving Layer-2 separation.

5.1 VLAN 10 — Employee

VLAN 10 is named EMPLOYEE and uses subnet 10.10.10.0/24 with default gateway 10.10.10.1. EMP-PC1 through EMP-PC5 belong to this VLAN. Employee users receive normal branch connectivity, including access to the internal server and external network, subject to the general routing and NAT configuration.

5.2 VLAN 20 — Guest

VLAN 20 is named GUEST and uses subnet 10.10.20.0/24 with default gateway 10.10.20.1. AP-GUEST extends this VLAN to GUEST-LAP1 and GUEST-LAP2. Guest traffic is intentionally treated as untrusted: it is allowed toward the external network but denied toward Employee, Server, and Management networks.

5.3 VLAN 30 — Server

VLAN 30 is named SERVER and uses subnet 10.10.30.0/24 with default gateway 10.10.30.1. SRV1 is statically addressed at 10.10.30.10. Keeping the internal server in its own VLAN creates a clear policy boundary and avoids placing shared services directly inside a user subnet.

5.4 VLAN 99 — Management

VLAN 99 is named MANAGEMENT and uses subnet 10.10.99.0/24 with default gateway 10.10.99.1. SW1 is managed at 10.10.99.2, SW2 at 10.10.99.3, and ADMIN-PC at 10.10.99.10. The management subnet is not intended for Guest access and is the trusted source area for remote administration.

6. IP Addressing Plan

Device / Scope	Interface / VLAN	IPv4 Address / Range	Mask	Assignment
R1-BRANCH	G0/0.10	10.10.10.1	/24	Static gateway
R1-BRANCH	G0/0.20	10.10.20.1	/24	Static gateway
R1-BRANCH	G0/0.30	10.10.30.1	/24	Static gateway
R1-BRANCH	G0/0.99	10.10.99.1	/24	Static gateway
R1-BRANCH	G0/1	203.0.113.1	/24	Static external interface
SW1	VLAN 99 SVI	10.10.99.2	/24	Static management
SW2	VLAN 99 SVI	10.10.99.3	/24	Static management
ADMIN-PC	VLAN 99	10.10.99.10	/24	Static management host
SRV1	VLAN 30	10.10.30.10	/24	Static server / DNS host
INTERNET-SRV	External	203.0.113.10	/24	Static simulated Internet host
Employee clients	VLAN 10	10.10.10.11-10.10.10.254	/24	DHCP-eligible; .1-.10 reserved
Guest clients	VLAN 20	10.10.20.11-10.10.20.254	/24	DHCP-eligible; .1-.10 reserved
Server subnet clients	VLAN 30	10.10.30.11-10.10.30.254	/24	DHCP-eligible if used; infrastructure remains reserved
Management subnet clients	VLAN 99	10.10.99.11-10.10.99.254	/24	DHCP-eligible if used; infrastructure remains reserved

Address reservation: The first ten addresses of each internal /24 are reserved from DHCP to prevent collisions with gateways and static infrastructure. The assurance requirements explicitly validate Employee and Guest DHCP default gateways.

7. Port Mapping

The design relies on three trunk links whose interface identities are part of the requirement baseline. Access-port numbers are not required by the Python assurance file, so this document maps endpoint groups by role/VLAN rather than inventing FastEthernet port numbers that are not part of the validated requirement set.

Device	Interface / Link	Connected To	Mode	VLAN / Purpose
R1-BRANCH	G0/0	SW1 Gi0/1	802.1Q trunk	Carries VLANs 10, 20, 30, 99 to router subinterfaces
SW1	Gi0/1	R1-BRANCH G0/0	802.1Q trunk	Allowed VLANs 10,20,30,99
SW1	Gi0/2	SW2 Gi0/1	802.1Q trunk	Allowed VLANs 10,20,30,99
SW2	Gi0/1	SW1 Gi0/2	802.1Q trunk	Allowed VLANs 10,20,30,99
R1-BRANCH	G0/1	INTERNET-SRV network	Routed	203.0.113.0/24; NAT outside
SW1	Access ports	EMP-PC1, EMP-PC2, EMP-PC3	Access	VLAN 10 EMPLOYEE
SW1	Access port	SRV1	Access	VLAN 30 SERVER
SW1	Access port	ADMIN-PC	Access	VLAN 99 MANAGEMENT
SW2	Access ports	EMP-PC4, EMP-PC5	Access	VLAN 10 EMPLOYEE
SW2	Access port	AP-GUEST	Access	VLAN 20 GUEST
AP-GUEST	Wireless association	GUEST-LAP1, GUEST-LAP2	Wireless	Guest clients bridged into VLAN 20

8. Inter-VLAN Routing

8.1 Router-on-a-Stick

SmartBranch 360 uses router-on-a-stick because the 2960 access switches operate at Layer 2 and the branch needs a single routed point between VLANs. R1-BRANCH uses one physical interface toward SW1 and divides it into four subinterfaces. Each subinterface terminates one VLAN, owns that VLAN's default-gateway address, and participates in routing and NAT policy.

This approach is appropriate for the project scale because it demonstrates VLAN trunking and inter-VLAN routing without requiring a multilayer switch. The main dependency is the router-to-switch trunk: if a VLAN is missing from the trunk or the matching subinterface is misconfigured, that VLAN loses routed connectivity.

8.2 802.1Q subinterfaces

Router Subinterface	802.1Q VLAN	IPv4 Address	NAT Role	Function
G0/0.10	10	10.10.10.1/24	inside	Employee default gateway
G0/0.20	20	10.10.20.1/24	inside	Guest default gateway; ACL 100 inbound
G0/0.30	30	10.10.30.1/24	inside	Server default gateway
G0/0.99	99	10.10.99.1/24	inside	Management default gateway
G0/1	-	203.0.113.1/24	outside	External / simulated Internet edge

```

interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.10.10.1 255.255.255.0
 ip nat inside
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.10.20.1 255.255.255.0
 ip nat inside
 ip access-group 100 in

```

Representative router-on-a-stick configuration pattern. VLAN 30 and VLAN 99 follow the same 802.1Q structure.

9. Network Services

9.1 DHCP

R1-BRANCH provides DHCP so client devices can receive addressing parameters automatically. The project configuration uses role-based pools named EMPLOYEE_POOL, GUEST_POOL, SERVER_POOL, and MANAGEMENT_POOL. The first ten addresses in each internal subnet are excluded so that gateways and reserved infrastructure addresses remain stable. Employee and Guest DHCP behavior is directly checked by the Python assurance tool because these are the principal dynamic client networks.

Pool / Role	Network	Default Router	DNS Server	Notes
EMPLOYEE_POOL	10.10.10.0/24	10.10.10.1	10.10.30.10	Employee clients
GUEST_POOL	10.10.20.0/24	10.10.20.1	10.10.30.10	Guest wireless clients
SERVER_POOL	10.10.30.0/24	10.10.30.1	10.10.30.10	Available for non-reserved server-subnet hosts
MANAGEMENT_POOL	10.10.99.0/24	10.10.99.1	10.10.30.10	Infrastructure remains statically reserved

9.2 DNS

SRV1 at 10.10.30.10 provides the internal DNS service used by branch clients. DHCP supplies 10.10.30.10 as the DNS server address. DNS is also treated as a distinct service during troubleshooting: a client can have working IP routing to an external address while name resolution fails if UDP/53 is accidentally blocked. This distinction is useful because it separates transport reachability from application-layer name resolution.

Design implication: Because Guest clients use an internal DNS service but are otherwise isolated from the Server VLAN, the Guest security policy must preserve the DNS behavior required by the final configuration while still blocking unauthorized general access to internal networks.

9.3 Guest Wi-Fi

AP-GUEST is attached to SW2 as an access connection in VLAN 20. GUEST-LAP1 and GUEST-LAP2 associate wirelessly with the AP and enter the branch through the Guest VLAN. They obtain Guest addressing, use 10.10.20.1 as the default gateway, and are subject to the Guest ACL on R1-BRANCH before reaching other routed networks.

9.4 NAT/PAT

PAT allows multiple private branch addresses to share the address of R1-BRANCH G0/1 when traffic is translated toward the external network. All four internal router subinterfaces are marked as NAT inside, while G0/1 is marked as NAT outside. NAT ACL 1 identifies the four internal subnets that are eligible for translation, and overload is performed on the G0/1 interface address.

```
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

ACL 1 permits:

10.10.10.0/24 (Employee)

10.10.20.0/24 (Guest)

10.10.30.0/24 (Server)

10.10.99.0/24 (Management)

During the revised NAT fault scenario, G0/1 intentionally loses its ip nat outside role and translations are cleared. Since INTERNET-SRV is directly connected to R1-BRANCH in the simulation, a ping can still succeed without proving PAT is healthy. The definitive evidence is therefore the presence or absence of a new translation in show ip nat translations after fresh traffic is generated.

10. Security Architecture

10.1 Guest Isolation

Guest isolation is enforced at the router, where inter-VLAN routing occurs. Traffic entering from VLAN 20 is evaluated before it can reach protected branch networks. The intended policy is simple and auditable: Guest traffic must not reach the Employee, Server, or Management zones, but it may reach the external network and the specific services required for normal guest operation.

Source	Destination	Policy	Reason
Guest VLAN 20	Employee VLAN 10	DENY	Protect employee endpoints from visitor traffic
Guest VLAN 20	Server VLAN 30	DENY by default	Protect internal services; only explicitly required service behavior is preserved
Guest VLAN 20	Management VLAN 99	DENY	Prevent access to administrative infrastructure
Guest VLAN 20	External network	ALLOW	Provide intended guest Internet access

10.2 ACL Policy

The implemented Guest policy is represented by ACL 100 and is applied inbound on R1-BRANCH G0/0.20. Applying the ACL inbound close to the Guest source prevents unauthorized traffic from being routed deeper into the branch. The requirement baseline and healthy assurance report both treat the presence and inbound application of this ACL as a mandatory security condition.

Fault testing deliberately introduces an ACL change that blocks Guest DNS. The resulting symptom is intentionally service-specific: direct IP connectivity can continue while name resolution fails. The repair removes the unintended DNS-deny behavior without weakening the Guest isolation policy.

10.3 Management VLAN

VLAN 99 separates network administration from user and guest traffic. R1-BRANCH uses 10.10.99.1 as the Management gateway, SW1 uses 10.10.99.2, SW2 uses 10.10.99.3, and ADMIN-PC uses 10.10.99.10. Switches use 10.10.99.1 as their management default gateway. Guest traffic is denied to this subnet.

10.4 SSH Security

Remote management is restricted by source rather than allowing unrestricted VTY access. The project security requirement specifies 10.10.99.10 as the only authorized SSH source. VTY lines are configured to accept SSH and use an inbound access-class that permits the authorized management host without additional permit sources. The Python assurance checker parses VTY blocks, validates the inbound access-class, and fails the check if the expected source is missing or extra sources are permitted.

11. Testing Strategy

Testing is organized by requirement rather than by device. A test is considered complete only when the expected endpoint behavior and the supporting device evidence agree. Both positive tests (traffic should succeed) and negative tests (traffic should be denied) are required.

Test ID	Test / Source	Expected Result	Evidence
T01	Employee DHCP	Employee PC receives VLAN 10 address and gateway 10.10.10.1	ipconfig / endpoint IP settings; router DHCP config
T02	Employee to SRV1	Employee reaches 10.10.30.10	ping / Packet Tracer simulation
T03	Employee external access	Employee reaches 203.0.113.10	ping plus NAT translation when generated
T04	Guest DHCP / Wi-Fi	Guest laptop associates and receives VLAN 20 addressing	wireless association and IP configuration
T05	Guest external access	Guest reaches allowed external destination	ping / service test
T06	Guest to internal zones	Guest cannot access Employee, Server (except explicitly required service behavior), or Management	failed ping/service evidence plus ACL placement
T07	Management SSH	ADMIN-PC 10.10.99.10 can SSH to network devices	SSH session / VTY configuration
T08	Unauthorized SSH	Non-management hosts are rejected for device management	failed SSH attempt / access-class evidence
T09	VLAN/trunk state	VLANs 10,20,30,99 exist and cross intended trunks	show vlan brief; show interfaces trunk
T10	Router baseline	Gateway subinterfaces, NAT roles, ACL, and DHCP settings match design	show ip interface brief; show running-config

12. Troubleshooting Strategy

Troubleshooting follows a repeatable evidence-first workflow: establish the healthy baseline, introduce one controlled fault, observe the symptom, identify the affected layer or service, capture a show-command or endpoint test that proves the root cause, repair only the faulty configuration, and then repeat the original test. Each fault is isolated so the evidence remains attributable to a single cause.

Fault	Injected Error	Primary Symptom / Evidence	Corrective Action
F01 - Wrong Employee default gateway	Employee gateway changed to 10.10.10.254	Local addressing exists but remote VLAN/external access fails	Restore Employee default gateway to 10.10.10.1 and renew/retest
F02 - VLAN 20 missing on trunk	SW1-SW2 trunk allows 10,30,99 instead of 10,20,30,99	Guest connectivity fails; checker reports VLAN 20 missing from SW1 Gi0/2	Add VLAN 20 back to the allowed trunk VLAN list
F03 - Wrong Guest DHCP gateway	Guest DHCP default-router changed to 10.10.20.254	Guest receives an IP but cannot route outside its subnet	Restore Guest DHCP default-router 10.10.20.1 and renew client
F04 - ACL blocks Guest DNS	Temporary ACL behavior blocks UDP/53	Direct IP access can work while DNS name resolution fails	Remove the unintended DNS block while preserving Guest isolation
F05 - NAT outside missing	Remove ip nat outside from R1 G0/1 and clear translations	Fresh traffic produces no proper NAT translation; ping alone is not definitive in this directly connected simulation	Restore ip nat outside, clear translations, generate fresh traffic, verify show ip nat translations

Fault 5 verification rule: Do not treat a successful ping to 203.0.113.10 as proof that NAT is working. INTERNET-SRV is directly connected to R1-BRANCH, so NAT health is verified from translation evidence after fresh traffic.

13. Python Network Assurance

The Python assurance component converts selected Cisco show-command captures into repeatable checks against requirements.yaml. This provides a second layer of validation beyond manual Packet Tracer observation. The tool is intentionally lightweight: it does not configure devices or replace troubleshooting; it detects mismatches and reports readable findings that direct attention to likely configuration errors.

Input	Purpose
requirements/requirements.yaml	Defines site name, VLANs, subnets, gateways, router interfaces, NAT roles, switch management addresses, expected trunks, server addresses, DHCP gateway requirements, Guest policy, and allowed SSH source.
SW1 show vlan brief	Confirms required VLAN IDs/names exist on SW1.
SW2 show vlan brief	Confirms required VLAN IDs/names exist on SW2.
SW1 show interfaces trunk	Validates expected trunks and allowed VLANs.
SW2 show interfaces trunk	Validates expected trunks and allowed VLANs.
R1 show ip interface brief	Validates gateway and external interface IPv4 addresses.
R1 show running-config	Validates NAT roles, DHCP default gateways, Guest ACL application, and SSH source restriction.

The checker normalizes interface names, parses VLAN and trunk tables, extracts router interface addresses, examines interface configuration blocks for NAT roles, identifies DHCP pools and their default-router statements, checks ACL placement on the Guest subinterface, and validates VTY access-class restrictions. Findings are emitted as PASS, WARN, or FAIL; the overall result is FAIL if any mandatory check fails.

SmartBranch 360 Assurance Summary

```
=====
Checks passed: 25
```

```
Warnings:    0
```

```
Checks failed: 0
```

```
Overall: PASS
=====
```

The trunk-fault sample demonstrates the diagnostic value of the same tool: with VLAN 20 removed from SW1 Gi0/2, the report changes to 24 PASS / 0 WARN / 1 FAIL and identifies the missing Guest VLAN together with the expected and observed allowed-VLAN sets.

14. Results

The final healthy SmartBranch 360 baseline satisfies the central project requirements. Both switches contain VLANs 10, 20, 30, and 99; the three intended trunks permit all four VLANs; R1-BRANCH owns the correct four gateway addresses and 203.0.113.1 external address; all internal subinterfaces are configured as NAT inside while G0/1 is NAT outside; Employee and Guest DHCP gateways are correct; Guest ACL 100 is present inbound on G0/0.20; and SSH management is restricted to 10.10.99.10.

Result Area	Observed Final Status
VLAN segmentation	PASS - required VLANs present on SW1 and SW2
Trunking	PASS - SW1 Gi0/1, SW1 Gi0/2, and SW2 Gi0/1 carry VLANs 10,20,30,99
Router gateways	PASS - 10.10.10.1, 10.10.20.1, 10.10.30.1, 10.10.99.1
External interface	PASS - R1 G0/1 = 203.0.113.1/24
NAT interface roles	PASS - inside on VLAN subinterfaces; outside on G0/1
DHCP gateway checks	PASS - Employee 10.10.10.1; Guest 10.10.20.1
Guest ACL	PASS - ACL 100 exists and is applied inbound on G0/0.20
SSH restriction	PASS - authorized source limited to 10.10.99.10
Python assurance	PASS - 25 passed, 0 warnings, 0 failed
Fault-detection example	PASS - VLAN 20 trunk fault detected as a specific FAIL finding

The troubleshooting exercise also demonstrates that the design is diagnosable rather than merely functional. Different failures produce distinguishable symptoms: routing failure from a wrong gateway, Layer-2 path failure from a missing trunk VLAN, control-plane addressing failure from a bad DHCP option, application-layer failure from DNS blocking, and translation-state failure from a missing NAT outside role.

15. Conclusion

SmartBranch 360 demonstrates a complete small-branch network lifecycle: plan the topology, segment users with VLANs, route between subnets, provide addressing and name services, connect wireless guests, translate private addresses for external access, enforce security policy, restrict management, test normal operation, inject controlled faults, and verify the final state with automation.

The final design is intentionally simple enough to understand in Packet Tracer while still reflecting important real networking principles. Functional separation is achieved through VLANs; routing is centralized on R1-BRANCH; Guest access is constrained by policy; management traffic is isolated and source-restricted; and

configuration assurance is made repeatable through Python. The result is not only a working topology but a documented, testable, and troubleshootable branch-office design.

Final design state: Healthy assurance report: 25 checks passed, 0 warnings, 0 failures. The saved topology, requirements file, troubleshooting evidence, and Python checker together provide the evidence chain for the SmartBranch 360 implementation.

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